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* Proposal dibawah ini adalah hasil diskusi dengan Gemini AI. Berikut saya lampirkan link percakapan: <https://gemini.google.com/share/382240a4c68f>

SummitSearch — A Multimodal Semantic Engine for Mountain Discovery

1. Overview of the System

SummitSearch is an intelligent information retrieval (IR) system designed for hikers and travelers who seek destinations based on "environmental vibes" and "characteristics" rather than just names or locations.

While traditional search engines rely on keyword matching (e.g., searching "mountains with snow"), **SummitSearch** understands the **semantic meaning** of a user's request. By combining detailed textual descriptions from *SummitPost* with visual data extracted from images, the system allows users to provide complex natural language queries like: *"A challenging, rocky peak with a volcanic atmosphere and cold climate."* The system doesn't just look for those words; it understands the "vibe" of the mountain through a hybrid approach of text and image analysis.

2. Core Algorithms & Models

The project utilizes state-of-the-art AI models to move beyond simple Boolean retrieval:

- **Sentence-BERT (SBERT):** Used for **Textual Embedding**. It converts long descriptions of mountains into dense vectors (mathematical representations of meaning). This allows for "Semantic Search" where synonyms (e.g., "chilly" and "cold") are treated as similar.

- **CLIP (Contrastive Language-Image Pre-training):** Used for **Visual Feature Extraction**. This "zero-shot" model will analyze mountain photos to automatically assign labels like "Tundra," "Forest," or "Glacial" without needing a manually labeled dataset.
 - **Cosine Similarity:** The primary **Ranking Algorithm**. It calculates the mathematical distance between the user's query vector and the mountain vectors in the database. The smaller the distance, the higher the rank.
 - **Lapse Rate Calculation:** A simple geographic heuristic used to estimate "Summit Temperature" based on base-level climate data and elevation.
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3. The Technical Pipeline

The system follows a six-stage pipeline from raw data to the final ranked list:

Phase I: Data Acquisition & Cleaning

- **Targeted Scraping:** Extracting names, descriptions, elevation, and image URLs from 200–500 high-quality entries on *SummitPost*.
- **Normalization:** Cleaning the text (removing HTML tags, non-ASCII characters) and resizing images for AI processing.

Phase II: Multimodal Indexing

- **Text Encoding:** The cleaned text is passed through the SBERT model to create a **Text Vector**.
- **Visual Tagging:** Images are passed through the CLIP model to extract "Vibe Tags" (e.g., Arid, Lush, Snowy).
- **Metadata Integration:** Altitude and regional temperature are stored as structured filters.

Phase III: Vector Storage

- All vectors and metadata are stored in a **Vector Database** (e.g., **ChromaDB** or **FAISS**). This allows for lightning-fast retrieval even as the dataset grows.

Phase IV: Query & Retrieval

1. **Input:** The user enters a natural language query.
2. **Vectorization:** The system converts the query into a search vector.
3. **Similarity Search:** The system retrieves the Top-K mountains that match the text meaning.

4. **Reranking:** The system "boosts" mountains whose visual tags (from Phase II) align with the query (e.g., if the user said "rocky," mountains with the "Arid/Rocky" visual tag move to the top).